THE UNIVERSITY OF SYDNEY

MOOC INTRODUCTION TO CALCULUS

## Notes for 'Parabolas and quadratics'

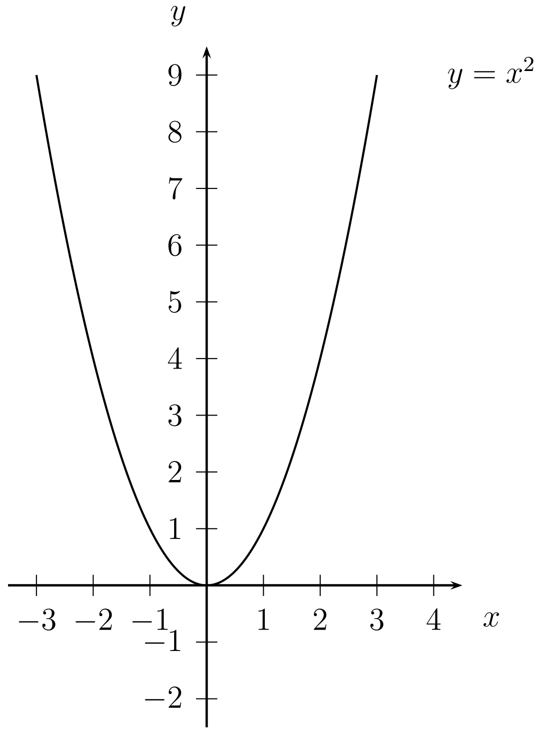
### Important Ideas and Useful Facts:

(i) Quadratic expressions: A *quadratic expression* (*in* $x$) has the form

$$ax^2 + bx + c$$

where $x$ is a variable and $a$, $b$ and $c$ are constants such that $a$ is nonzero. (If $a$ were zero, then the expression would collapse to $bx + c$, which is called a *linear expression*.)

(ii) Parabolas: A *parabola* is a curve in the plane that is the result of any combinations of (positive or negative) dilations, rotations or translations of the curve $y = x^2$.




This particular parabola contains, for example, the points $(0, 0)$, $(1, 1)$, $(-1, 1)$, $(2, 4)$ and $(-2, 4)$. The curve associated with any quadratic expression $y = ax^2 + bx + c$ is a parabola, *facing upwards* if $a > 0$, and *facing downwards* if $a < 0$.

(iii) Completing the square: The variable $x$ in the quadratic expression $ax^2 + bx$, where $a$ is nonzero, may be isolated by the technique of *completing the square*:

$$ax^2 + bx \;=\; a\left(x^2 + \frac{bx}{a}\right) \;=\; a\left(x^2 + \frac{bx}{a} + \frac{b^2}{4a^2} - \frac{b^2}{4a^2}\right) \;=\; a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a}\,.$$

For example,

$$2x^2 + 3x = 2\left(x^2 + \frac{3x}{2}\right) = 2\left(x^2 + \frac{3x}{2} + \frac{9}{16} - \frac{9}{16}\right) = 2\left(x + \frac{3}{4}\right)^2 - \frac{9}{8}\,,$$

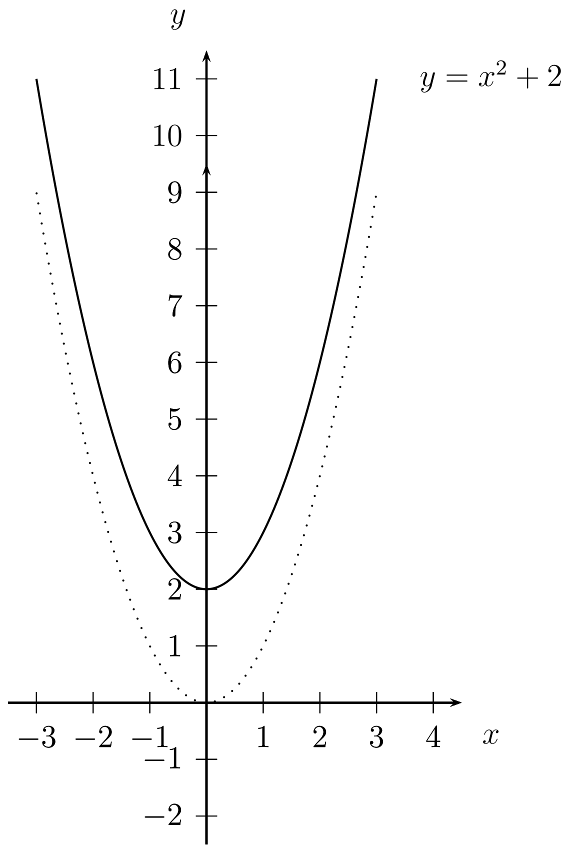
yielding only one occurrence of $x$.

**Examples:**

**1.** Consider the parabola defined by the quadratic expression

$$y \;=\; x^2 + 2\;.$$

This curve contains, for example, the points $(0,2)$, $(1,3)$, $(-1,3)$, $(2,6)$, $(-2,6)$, the result of adding 2 to the corresponding $y$-values of the simpler quadratic $y = x^2$.




The parabola is the result of shifting (translating) the parabola for $y = x^2$ vertically upwards by 2 units.

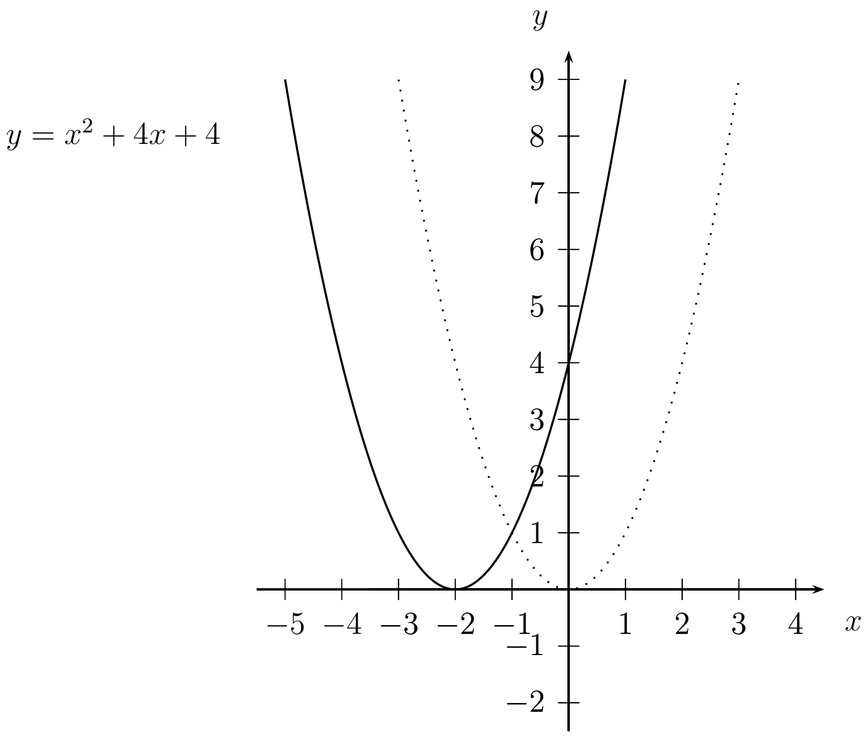
**2.** Consider the parabola defined by the quadratic expression

$$y \;=\; x^2 + 4x + 4\;.$$

This curve contains, for example, the points $(0,4)$, $(1,9)$, $(-1,1)$, $(2,16)$, $(-2,0)$. These points might appear haphazard compared to the previous examples, but in fact

$$y \;=\; x^2 + 4x + 4 \;=\; (x+2)^2\;,$$

so that the same $y$-values arise as for $y = x^2$, but by using inputs $x$ that are 2 units less in each case. Thus, the parabola is the result of shifting (translating) the parabola for $y = x^2$ horizontally to the left by 2 units.



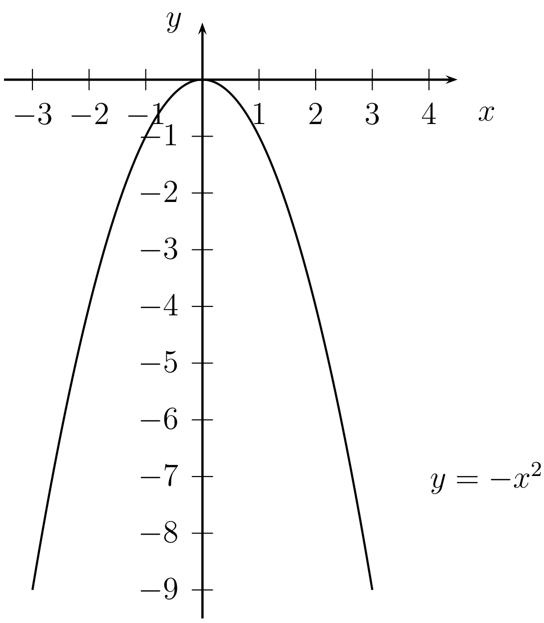

**3.** Consider the parabola defined by the quadratic expression

$$y \;=\; 2x - x^2 \;.$$

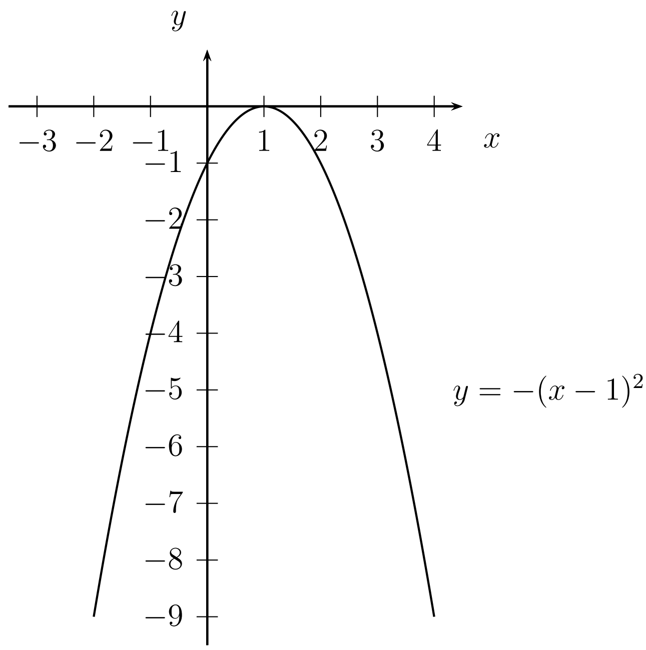
Because the coefficient of $x^2$ is negative, we expect the curve to be an upside-down parabola. To see exactly where it is positioned, we can use the trick of completing the square:

$y = 2x - x^2 = -(x^2 - 2x) = -(x^2 - 2x + 1 - 1) = -(x^2 - 2x + 1) + 1 = -(x-1)^2 + 1\;.$

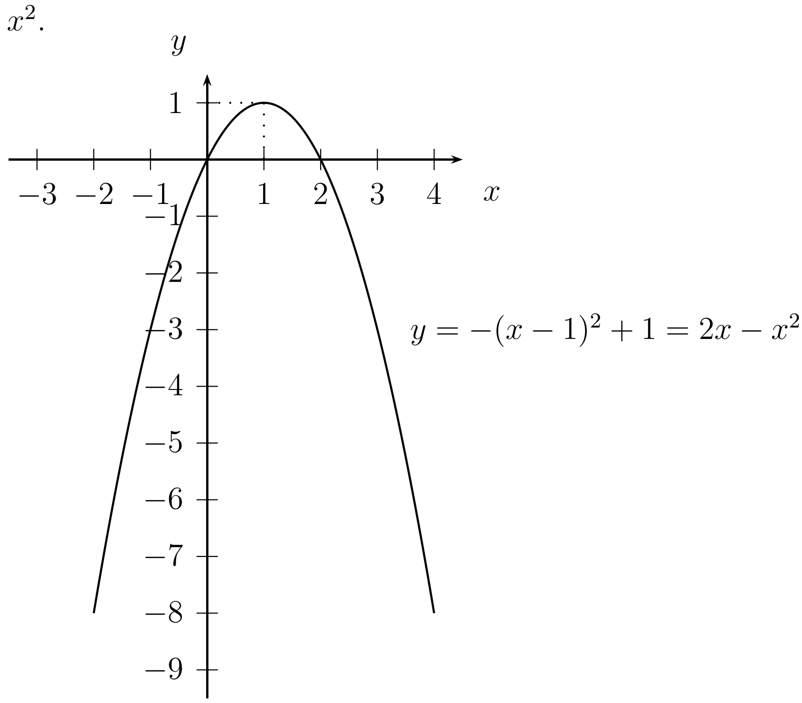
The curve $y = -(x-1)^2 + 1$ can be obtained from the parabola $y = x^2$ in several steps, firstly by "tipping it upside-down" by forming $y = -x^2$,



secondly, by shifting this curve one unit to the right, by replacing $x$ by $x-1$, to form $y = -(x-1)^2$,




and then, finally, translating this curve vertically upwards by one unit, to finally get $y = -(x-1)^2 + 1 = 2x - x^2$.




Notice how the apex of the parabola has shifted from the origin to the point $(1,1)$. Notice also that the final parabola crosses the $x$-axis at $x=0$ and $x=2$. In fact, an alternative way of locating the position of the final parabola is to realise that the quadratic factorises

$$2x - x^2 \;=\; -x(x-2)\ ,$$

which equals zero when one of the factors $x$ or $x-2$ is zero. Later, when we develop the *derivative*, we will also see a very fast way of locating the apex.

**4.** The shifting effects in the last few examples are special cases of a general phenomenon. Suppose that $f(x)$ is some expression involving $x$ (and later we place this in the wider formalism of *functions*), and $c$ is a positive constant. Then the curve $y = f(x)$ can be shifted vertically upwards $c$ units in the $xy$-plane by forming the curve $y = f(x)+c$. This makes sense, as we are simply adding $c$ units to all of the $y$-values.

To shift the curve downwards by $c$ units we just form the curve $y = f(x) - c$.

For horizontal shifts, the rules might look slightly unintuitive at first. To shift the curve $y = f(x)$ to the right by $c$ units, we form the curve $y = f(x - c)$. The reason for the minus sign is that we want to use inputs $x$ that are $c$ units larger in order to produce the same original $y$-values, and the effect is to translate the curve to the right.

To shift the curve $y = f(x)$ to the left by $c$ units, we form the curve $y = f(x + c)$. The reason for the plus sign is that we want to use inputs that are $c$ units smaller in order to produce the same original $y$-values, and the effect is to translate the curve to the left.

**5.** Consider a general quadratic expression

$$y \;=\; ax^2 + bx + c$$

where $a$, $b$ and $c$ are constants such that $a$ is nonzero. By the method of completing the square we obtain

$$y \;=\; ax^2 + bx + c \;=\; a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a} + c\,,$$

so

$$y \;=\; a(x + d)^2 + k$$

where $d = \frac{b}{2a}$ and $k = -\frac{b^2}{4a} + c$. Hence we can obtain the curve in three steps:

(1) firstly, translate the parabola $y = x^2$ to the left $d$ units (interpreted as a right translation if $d$ happens to be negative), to obtain the parabola $y = (x + d)^2$;

(2) secondly, dilate the parabola $y = (x + d)^2$ by a factor of $a$, to obtain the parabola $y = a(x + d)^2$, which involves a reflection in the $x$-axis if $a$ happens to be negative;

(3) thirdly, shift the parabola $y = a(x + d)^2$ vertically upwards by $k$ units (interpreted as downwards if $k$ happens to be negative), to finally produce the parabola $y = a(x + d)^2 + k$.

During this process we have only ever produced translations or dilations of parabolas. This explains why the curve of a general quadratic expression is always a parabola. The explanation includes, in fact, a method or procedure for locating precisely where the final parabola should sit in the $xy$-plane, based on the method of completing the square.